Ashwin Khandelwal

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Education

Delhi Technological University — New Delhi, India
B.Tech in Computer Science & Engineering — CGPA: 8.48/10 — *2023 – 2027*

Delhi Police Public School — New Delhi
CBSE (Class XII) | Aggregate: 91.4% — *2023*
CBSE (Class X) | Aggregate: 95.8% — *2021*

Experience

Software Developer Intern — May 2026 – Jul. 2026
JDMR iSchool — *Gurgram, Haryana*

- **Architected** a multi-layer AI guardrail pipeline for a conversational school ERP assistant, applying **AWS Bedrock's ApplyGuardrail** and **InovokeGuardrailChecks API** for content-policy violations, prompt-attack, and jailbreak patterns paired with hallucinations and contextual grounding along with Automated Reasoning.
- **Reduced** guardrail-processing cost by **front-loading** the InvokeGuardrailChecks API ahead of ApplyGuardrail's full policy pipeline — reducing cost per 1,000 text units from **$0.40 on every request** to **$0.15 on blocked/malicious requests** (62.5% reduction) and **$0.25 on legitimate requests** (37.5% reduction)
- **Monitored** guardrail activity via **AWS CloudTrail** logging to track blocked/flagged requests, using those insights to **iteratively tune custom regex** and word-filter rules for better detection accuracy.

Projects

TrueDueAI | *Python, Langchain, ChromaDB, all-MiniLM-L6-v2, StreamlitUI* | GitHub — June 2026

- Built a RAG-based Financial Due Diligence AI assistant for financial analysts to upload financial documents and ask due-diligence questions via a chat interface, backed by a **provider-agnostic LLM layer(via Ollama - LLama(Groq) - DeepSeek - Qwen)** with automatic failover across LLMs
- Engineered a **multi-tool ReAct Agent** with autonomous tool selection processing queries at **3.42s average response time** tracked via console based metrics dashboard.
- Implemented semantic retrieval over a ChromaDB vector store, wired into a citation-grounded RAG chat pipeline with deterministic financial-ratio calculations and Pydantic-validated extraction to prevent hallucination

Automated Info Management System | *MERN Stack* | GitHub — June 2025

- Developed a full-stack request-and-approval based faculty advisor selection system for mentorship and course registration, handling **100+** records across **5+** courses
- Implemented RBAC with JWT authentication, blocking unauthorized access attempts
- Optimized MongoDB schema with a combination of simple and compound indexes, cutting query latency by **62% (120ms → 45ms)** and improving dashboard responsiveness for **50+** student records

Codeverse | *React, Socket.IO, Node.js* | GitHub — December 2025

- Built a collaborative code editor supporting **C++, Java, Python, and JavaScript**, enabling live multiuser coding sessions
- Engineered a scalable **Socket.IO/WebSockets** architecture handling more than **500** live code edits per minute, with instant synchronization and typing indicators

Technical Skills

Programming: C++, Python, JavaScript
Web: HTML, CSS, ReactJS, NodeJS, ExpressJS, StreamlitUI
Databases: MySQL, PostgreSQL, MongoDB, ChromaDB (Vector-based)
Tools: Git, GitHub, WebSocket, pdfplumber, Langchain
Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Generative AI, Agentic AI, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

Achievements & Activities

Hackathons — Secured top positions in HackCBS 8.0, HackVSIT 6.0, and brAInwave
Competitive Programming — Solved 800+ problems across Codeforces/LeetCode/GfG; best rank in top 15%
Certifications — Graphs Camp (AlgoUniversity), AWS APAC Solutions Architecture (Forage)